Identification of bumblebee species based on their wings

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***Abstract*—Bumblebees play a vital role in maintaining ecological balance and agricultural productivity through their crucial role in pollination. However, the conservation of bumblebee populations presents challenges due to their cryptic nature, making species identification difficult without harming the bees. In this work, we propose a deep learning approach to predict bumblebee species instantaneously using wing images and landmark detection. We explore two spatial representations for the ground truth, focusing on 18 landmarks critical for species identification. Utilizing the U-Net architecture and data augmentation techniques, we train our model on a dataset of annotated wing images. New metrics are implemented to evaluate the model : correct prediction rate and mean distance between paired points. Our results demonstrate promising performance, with the proposed deep learning model achieving a correct prediction rate of up to** 0.77 **and a mean distance of 1.21 pixels between paired points on annotated wing images. We discuss potential improvements, including transfer learning, variability in dataset quality, and integration with morphometric analysis for comprehensive species identification. This work contributes to the development of efficient and accurate methods for bumblebee conservation and ecological research.**

I. Introduction

Bumblebee holds a vital role in maintaining ecological balance and supporting agricultural productivity. As a species crucial for the environment, bumblebees contribute significantly to pollination, ensuring the reproduction of various plant species and the sustainability of ecosystems.

However, the complexity of bumblebee populations presents challenges for their conservation. Bumblebees are cryptic species, meaning they display diverse genetic characteristics within a single species, making identification more difficult. For some species, bumblebees need to be anesthetized before being examined by a scientific expert in a laboratory to identify from which species they belong to. Even if knew anesthesia methods are being implemented [1], most of the time bees do not survive.

The inability to identify species on the ground, without killing them, raises challenges for the census and the protection of these crucial pollinators. To help scientists for the identification of these species, our idea was to use deep learning algorithms to predict instantaneously species based on bumblebee wings pictures which could be taken on the ground. Wings being unique and immutable species indicators for bees, complex biometric algorithm such as an adapted FaceNet for bumblebees could have been implemented for this work.

However, we decided to draw our inspiration from what is done currently by scientists. They annotate 18 landmarks corresponding to the veins intersections on the wings. The annotation is not randomly done at an intersection : the landmarks is annotated at the same relative position at the junction. This technique depends on specific conventions and have to be the same for all the studied wings.

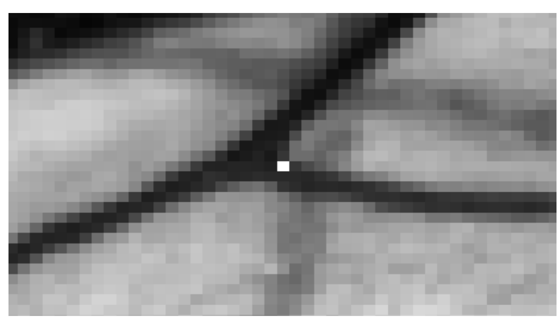

Fig. 1. Example of landmarks annotations on veins intersection

Then, using a morphometric software, scientists can predict the species bumblebees belongs to. [2]

Detecting the 18 landmarks on a bumblebee wing suffice to identify it. Reducing the complexity of identification algorithm, this proven method is our objective for the rest of the study given the accessibility to dataset suitable for this work.

II. Related Work

Some applications existed for bees identification based on wings images such as the Deep Automated Bee identification system. [3] This software offers accurate results and is downloadable on smartphone for a direct use on the ground. However, some species are more complex than others, their morphological similarities raise challenges for their identification. In the case of the lizards, tests were done to check identification for such species. [4] Results are encouraging, experts work could be lightened for the identification of cryptic species. Computer vision has a pivotal role to play in the scientific community.

Concerning bees identification, experts had to collect the 18 landmarks from wings which is tedious and time-consuming before loading the coordinates in a morphometric algorithm identifying the species. The morphometric analysis is based on the comparison of different conformations coming from the 18 landmarks of different wings from different species. This technique is explained in Adrien PERRARD's Thesis. [5]

The objective of the project explained in this paper is to replace the landmarks collection task and free time to scientist for relevant work.

III. Approach

A. Datasets Overview

Two datasets were at our disposal: the first from the Institute of Mons of 1700 pictures of bumblebees wings with the 18 corresponding points , the second from the IEES (the Institute of Ecology and Environmental Sciences of Paris) of 2500 pictures of bumblebees wings belonging to different species but without annotation. Therefore, with these 18 landmarks, the first dataset has been set for the learning, the validation and the test phase.

The final objective is to predict species from pictures of the second dataset which has a specific and precise way of taking pictures in field : wings open on an opaque paper, always in the same orientation. Conversely, the pictures from the first dataset have been taken in a laboratory on a transparent paper which gives better contour of veins. The orientation of wings also varies over the first dataset which can help the algorithm to generalize during the learning stage but make it more difficult for the algorithm to learn the pattern. Moreover, there is a grid in the background which does not appear in the second dataset.

To prepare the images for the learning, we decided to get closer to the pictures of the second dataset being the future tested images. We did a manual sorting taking out bees with two wings reducing the initial dataset to 1586 images. Then, thanks to the given 18 landmarks, we have been able to identify the wings orientation and to apply the adapted transformation to obtain the wing as follow. To identify the

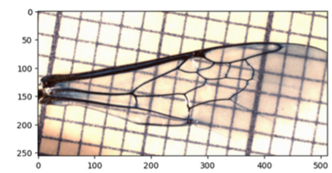

Fig. 2. Wing from Mons dataset after preprocessing

orientation, we calculate the lonely points of the landmarks which should always be the one with the highest coordinates (at the top right of the picture). This condition was sufficient to know if we flip vertically or/and horizontally the picture.

Finally, the picture was reshaped to fit the network architecture.

B. Ground Truths

The ground truth, the 18 landmarks of the wings, can be described differently to the network. The coordinates could be use as the label of the image, but it does not make the most of it. Given the small dataset, the more information we can transfer to the network through labels, the better it will learn, the best will be the prediction. Even if it needs a point recovery algorithm, a spatial representation of the points over a picture could help the learning process.

This has already been implemented to count the number of objects in a picture [6] using the integration of discrete normalized gaussian points representation distributed over the picture. However, even if the normalization gives information regarding the number of points, it does not specify accurately where this point is, especially in the case of two close points in which the summation of the gaussian leads to a pic in between the points. Once the gaussian output is given, the collection of the points raises challenges.

To circumvent it, we put the normalisation aside and implemented a sharp curve around the landmarks. We defines the following variables :

- the 18 images corresponding each to a unique landmark $(I_k)_{1\leq k\leq 18}$;
- the coordinates of the k^{th} landmark (x_k, y_k);
- the distance from the landmark where pixels are transformed r.

Then, we can obtain the wanted ground truth :

$$I = max((I_k)_{1\leq k\leq 18})$$

I_k is defined as follow

$$If \quad |i - x_k| > r \ or \quad |j - y_k| > r,$$

$$I_{kij} = 0$$

$$else,$$

$$I_{kij} = \frac{(r - \sqrt{(i-x_k)^2 + (j-y_k)^2})^2}{r^2}$$

A continuous landscape with pics equal to one at the landmarks coordinates and surrounding by decreasing numbers with a quadratic slope. Empirically, r is set to 80 it was a trade-off between calculation or memory cost and a non-black screen network output.

Concerning the shape of the function surrounding landmarks, it was as steep as possible to get precise pics in the network output and ease the landmarks collection. It was balanced by the need of information in the ground truth that the network does not accept black screen as a valid output and no learning occurs.

These trade-offs were challenging, and I add bad results for this representation as explained later in the fourth part $ResultsandPerspective$. To take advantage of this spatial landmarks representation, I have to distinguish the landmarks. Some points are close one to another, and in the case of smooth pic, there is a confusion between the landmarks for the network and only one pic remains in the output. In fact, for some wings the closest points are only separated by 3 pixels.

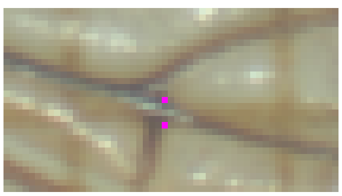

Fig. 3. Zoom on two close landmarks (red dots) on a bee wing

Two pairs of points are particularly challenging. Therefore, the slope must be as steep as possible for these points in order to isolate the pic accurately and have a chance to detect them. Conversely, other points are spread out over the image. The steep of the slope is not primordial to detect them. These points remain crucial to help the algorithm to converge to a decent result. In fact, the less steep the slope is the furthest the ground truth is from a black image.

Thus, an adapted slope for each point could give better results. The adapted ground truth is a transformation of the last one with steepest slope for the two challenging pairs of points in the detection due to their close distance. These landmarks are simply squared compared to the late ground truth.

C. Landmarks Recovery

In the output, the landmarks are represented by the highest values. Given the noisy results, the local maxima do not represent the landmarks. A filter needs to be applied to keep a small disk around the landmarks to avoid unwanted detection. For the closest points the disks are sometimes joined, and it has more the shape of a peanut. Calculating their centroid to determine the landmarks would consequently not be robust Thus, we collected the h-maxima of the filtered image and optimized h by dichotomy to obtain the 18 landmarks for each image.

A h-maximum point is both a local maximum of the image and a local maximum of the h-transform image. The h–transformed image is defined as follow. For a local maximum of level H, all its surrounding points higher than H-h are reduced to the value H-h or to the value of the higher local minimum in the surrounding area if local minimum value has an higher value than H-h. The surrounding are is defined as the points reachable by a path of points of constant or decreasing value from the local maximum. The h-transform image can be defined as "all mountain pics are flattened" as described by the following picture :

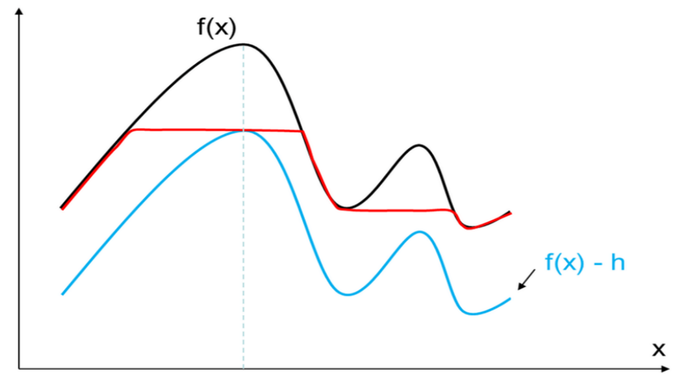

Fig. 4. H-transform result from the function f [9]

In the function above, only one point is defined as a h-maximum. If the network output discriminates the closest points by two different pics, the h-maxima algorithm can detect the points. However, if the output lack of accuracy landmarks cannot be detected. If a point is not predicted by the network, applying a filter prevents most of the time the h-maxima algorithm to return a maxima coming from the noisy output due to randomness in the network. This reduces the case of false positive landmarks and only 17 or even 16 points are recovered from the output. This way, morphometric analyses being a comparison of conformation is still possible by deleting the missing points for all the predictions. This results in a more robust output. However, the network should be trained, optimized and perfected to avoid the restricted case of 17 or 16 predicted points which reduces morphometric identification precision.

D. Hyperparamters for a U-Net architecture

The classical architecture for Deep Convolutional Neural Network for image analysis that we used is the U-Net architecture [7]. The inputs shapes were defined to fit the architecture while being compatible with data : a multiple of 32 was necessary to deal with the four contraction phases. With a four level encoder and an input shape of 256x512x3, a domain of 16x32x256 dimensions was obtained. 16 was the set number of filter to reach the maximum capacity available given the dataset. To train our U-Net model, the dataset was shuffled and split into sets of 1332 train, 148 validations, 106 test instances.

To avoid overfitting and ease the generalisation data augmentation was added to the learning process. For each epoch, data augmentation was set randomly in each batch of 32 images. The parameters for rotation, horizontal and vertical shift,

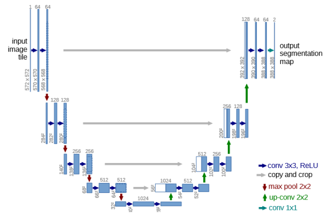

Fig. 5. U-Net architecture [6]

zoom, brightness, sharpness, hue, saturation are respectively 10, 0.05, [0.92,1.08], 0.2,1,0.2, 0.7. The previously described architecture is trained over 300 epoch with an Adam optimizer and a L1 loss (mean absolute error). All the training part and the model were coded using Tensorflow.

E. Metrics

The loss function was used on the validation dataset to evaluate the model. It gives the distance between the output and the ground truth, but it does not describe precisely if the recovered points are accurate or not. To define a metric for the points, we had to pair the predicted points to the ground truth points with the challenge of some unpredicted points, false negative and some predictions not associated to ground truth, false positive. Therefore, we associated the closest predictions to the ground truth points and if some ground truth had the same predicted points we kept the pair of points with the smallest distance. The others ground truth points were considered unpredicted and became false negative. Then, we checked if all the predicted points were associated to a ground truth, if not I collected them. This association is symmetric and bijective between the predicted and the ground truth points. The figure below help to represent the matching system.

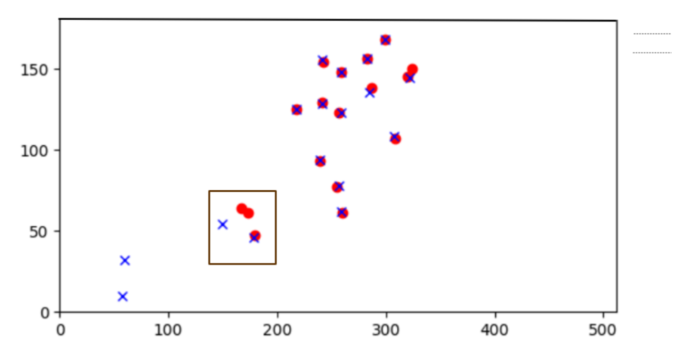

Fig. 6. The superposition of the ground truth landmarks (red dot) and the detected points (blue cross)

Looking at the frame inside the figure above, we can see one predicted point on a ground truth landmark which will be paired with and two ground truth points with a predicted point further. If this point is closer than five pixels to one of the ground truth it will be paired with the closer one and the algorithm avoid the last ground truth to be associated with the noisy detection at the bottom left of the picture (being further than five pixels). Else, the two points are not predicted and said false negative.

Based on this information, we define the accurate rate which is the rate of wings from which the network predicts the 18 paired landmarks with a distance lower than 5 pixels and with no false negative and no false positive prediction. It can be associated to the rate of correct network output. I also calculate the mean distance between paired points to have an idea of the accuracy of the predicted points.

IV. Training and Results

A. Homogeneous Ground Truth

For the first ground truth spatial representation, landmarks are described identically. The training curve is represented on the following graph.

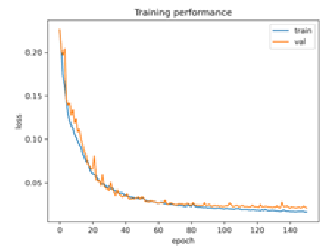

Fig. 7. First training curve

The learning curve decreases significantly during the first 30 epochs. The validation loss follows the training loss until the 100^{th} epochs. For the last 50 epochs, the curves stabilise but the decrease of the training loss and the stillness of the validation loss is a typical feature of overfitting. The best result is based on the smallest loss validation.

We obtain a validation loss of 0.0182, a correct rate of predicted wings of 0.49 and a mean distance for true positive points of 1.27 pixels.

The rate of good predictions is low. Even if the dataset is compound of wings with hidden landmarks, the rate should be increased. In fact, even images without obstruction are not predicted correctly because of close landmarks which have not been differentiated by the network. Value being close around the points, predicting one pic gives good results, explaining why the network did not try to modify this prediction during the learning.

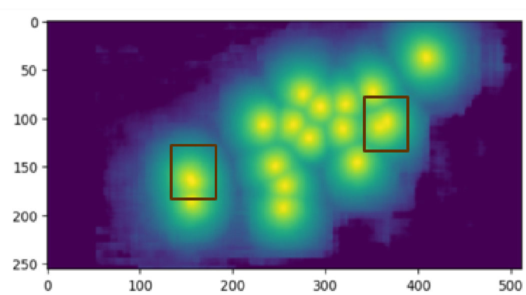

Fig. 8. A full output with framed pairs of closest landmarks

The figure 8 represents the output of the network for a wing from the test dataset. We can see in the frames that landmarks are hardly distinguishable.

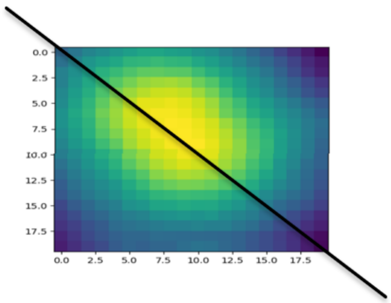

Fig. 9. A zoom on the left pair of landmarks

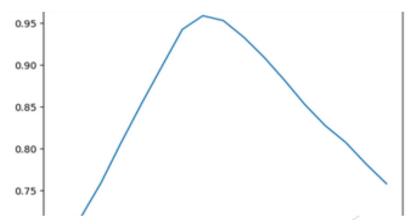

Fig. 10. The cross section of the picture above

The cross section of the zoom in the left frame emphasizes the impossibility to detect two landmarks in this area because there is only one maximum in the neighborhood.

However, if the landmarks are predicted, the results are accurate. This means that human points identification is learned by the algorithm because the diameters of knot veins can vary from about 4 to 12 pixels. A random prediction within the knots would result in a minimum distance of 2 pixels (the radius of the minimum circle), points being approximately situated at the rim of the inscribed circle of the intersections. This precision is emphasized by the fact that predictions are randomly and decreasingly distributed around the landmarks meaning no bias in the prediction as could be expected if the predictions were always detected at the middle of the intersections. The figure 11 represents the distribution for the first point but it looks roughly the same for all the landmarks.

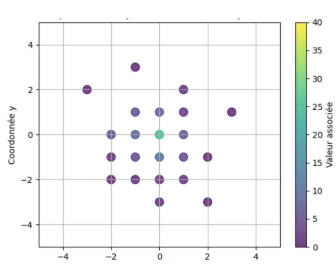

Fig. 11. Distribution of predictions centered around the ground truth for the 1^{st} landmark

B. Specific Ground Truth

To upgrade the algorithm, the learning is this time based on the second ground truth spatial representation with steeper variations around the two pairs of close points. The initial weights are based on the previous experienced weights. The training curve is represented on the figure 12.

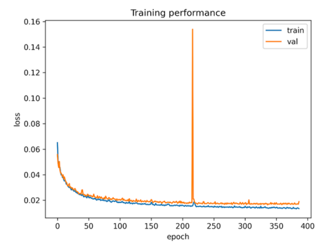

Fig. 12. The second training curve

Except an aberration for about the 210^{th} epochs, and some overfitting for the last epochs, the curve is looking great : the validation loss is decreasing and converging to a constant value, it is following the training loss. The best validation loss was 0.0160 which can not be compared to the previous experience given the ground truth change.

We obtain a correct rate of predicted wings of 0.77 and a mean distance for true positive points of 1.21 pixels. This model gives better results for all metrics : a higher rate of images with the 18 landmarks predicted and a lower average distance between the paired points.

Close points are better differentiated. However, even if some bad predictions result from the bad picture quality, some close points remain indistinguishable.

As explained before, there is the possibility to use only 17 or 16 points which would result in a better correct prediction rate. However, it is important to know if there is an interference between the closest points resulting in a deterioration of points precision. To check this, the distribution of predicted landmarks around the ground truth points is plotted below for both pair of close points when one of the two points is missing.

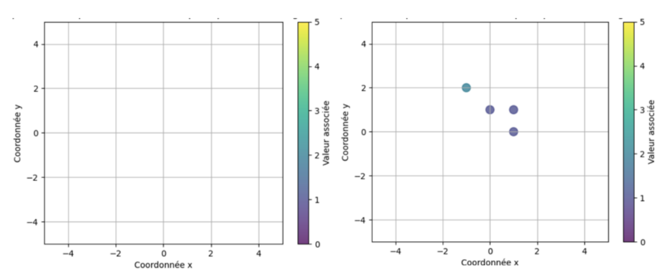

Fig. 13. Distribution of predictions centered around the ground truth for the 6^{th} (left) and 7^{th} (right) landmarks when missing the prediction for the 6^{th} or the 7^{th} points

For both of the pair of points, one landmark is more dominant than the other : the 7^{th} and the 12^{th} points are always predicted (except 5 times for the 12) and the predictions

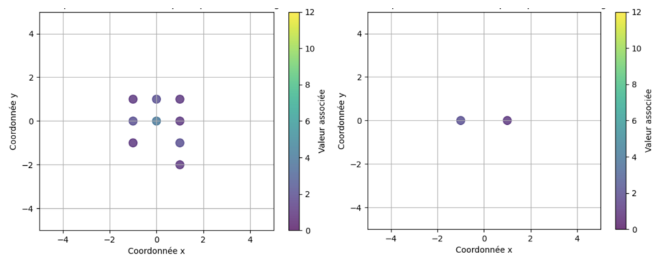

Fig. 14. Distribution of predictions centered around the ground truth for the 12^{th} (left) and 18^{th} (right) landmarks when missing the prediction for the 12^{th} or the 18^{th} points

stay really close to the ground truth. There is no aberration concerning the predictions of the points when the neighbor is missing which could have influenced the highest points of the area and consequently the predicted point.

To increase the scope of the model, reducing to 17 or 16 predictions is possible. However, the same point should be deleted for all the images. For only 17 landmarks needed, the point which should be forgotten is the 18 because it is the most undetected landmark in the model prediction. Taking it out, the rate of correct prediction increase to 0.83. Taking also the 6 out which is the one the less detected on the second pair of close points, the rate of correct prediction increase to 0.85. This expansion is clearly interesting for 17 points concerning the performance of the model. However, this should be check with the change in the capacity of identifying species based on morphometric analysis. The rate of correct predictions is lower with 17 than 18 landmarks but if the points are still precise enough with the model this can be compensated by the higher rate of good predictions by the model.

C. Test On Adriens Dataset

Images from this dataset do not resemble to the dataset from the University of Mons. However, no images being annotated, a transfer could not be done and the model could not be adapted to this dataset, we have to rest on its capacity of generalisation. The most problematic is the environment around the wing in Adrien's dataset. In the first dataset, all the picture are done on a grid which can be seen through the wings.

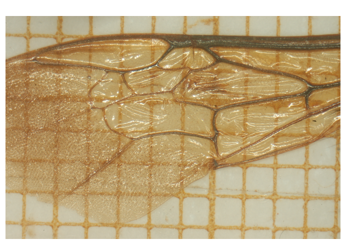

Fig. 15. Picture of a wing in Mons dataset

Conversely, in the second one, a grid pattern with black and white squares giving the scale is captured at the top.

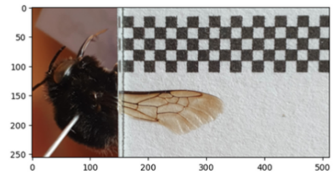

Fig. 16. A picture from Adrien's dataset

As was expected, the grid at the top is problematic. However, the model still manages to detect the wings and some points.

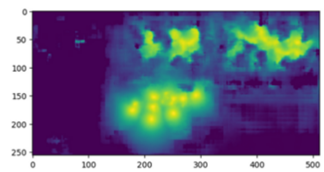

Fig. 17. Output from the model for the above input

To avoid the problem the picture was cropped that the wing fits perfectly the frame.This was done manually for dozens of wings from different species. This small sample represent the wide variety of species. To evaluate the results, I compare visually the output and the input to check that the pic are situated at the knot between the veins, no ground truth being available. For most of the species, the results are accurate. The model disregards the grid in the background and adapts to the brightness and contrast of these images. However, two species did not give the ground truth expected.

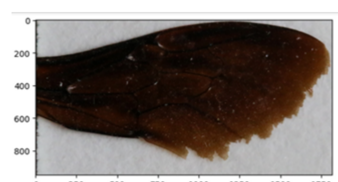

Fig. 18. Rupestris wing from Adrien's dataset

First, the species Rupestris returns a black images, no landmarks is detected.

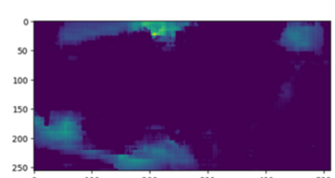

Fig. 19. Network output for the above Rupestris input

This is due to the thick paper on which the wing rests which is not transparent. This blocks the light and the wings is darker. In the case of a dark wings, the veins cannot be distinguish by the network. However, if only one specie is undetected, finally, we can identify it.

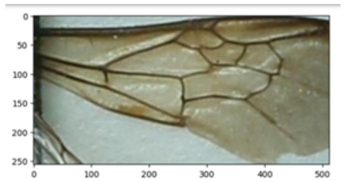

Fig. 20. Sichelli wing from Adrien's dataset

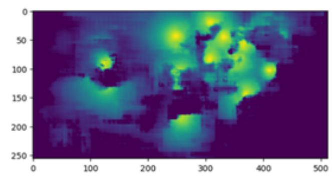

Fig. 21. Network output for the above Sichelli input

For the sichelli species, the output is inaccurate.

The output is noisy. This is due to the low quality of the picture which was taken with a different device. This can be corrected by specifying the photographer which device works and which one to use. The crop is close to the wings, border effects can also alter the output.

Thus, some errors exist for specific species and can be corrected. Overall, this results on Adrien's datset are promising.

V. Conclusions and Perspectives

To conclude, this research on points detection raises several challenges especially on the adapted ground truth to implement. The solution of a landform over the picture with slope adapted to the need gives promising results. However, there are many others possibilities to represent the ground truth. What might be even more efficient could be 18 output images, the $(I_k)_{1\leq k\leq 18}$ each corresponding to a landmark. This will help for the separation of the closest points and give directly the labeling of the points which could speed up the matching ground truth and predicted point phase.

For a complete analysis, the detection of species based on morphometric system, is needed. It will give details on the precision of the model and determine the number of landmarks to detect. The test will be all the more relevant as the pictures respect the meticulous procedure of photograph of bees on the ground. Therefore, the pictures will be cleaner and nothing alter the detection.

Moreover, a transfer should be done to upgrade the accuracy of the predictions. Images are really similar across the dataset, a transfer could be very beneficial and help the model for the predictions of some species such as the dark one. Taking high quality picture for all wings is feasible being the case for most of the images and could also improve the identification system as the Sichelli species problem shows.

The Weighted Hausdorff Distance metric [8] could give information about the accuracy of the point detection. Being differentiable, it could be implemented in the training phase and lead the network to an output adapted to the collecting technique to obtain more accurate results.

As a conclusion, this work offers promising perspective for the identification of cryptic species based on a deep neural network for image analysis.

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